Chapter 17, Problem 3RQ

Problem

Which of the following are even functions?

(a) *t* + *t*2

(b) *t*2cos *t*

(c) *et*2

(d) *t*2 + *t*4

(e) sinh *t*

Step-by-step solution

Step 1 of 5

(A) $f(t) = t + t^2$

$f(-t) = -t + (-t)^2 = -t + t^2$

$\boxed{f(t) \text{ is not even}}$

Step 2 of 5

(B) $f(t) = t^2 \cos t$

$f(-t) = (-t)^2 \cos(-t)$

$f(-t) = t^2 \cos t$

since

$\mathrm{f}(t) = tf(-t)$

$\boxed{f(t) is\ even}$

Step 3 of 5

(C) $f(t)e^{t^2}, f(-t) = e^{+t^2}$

$\boxed{f(t) is\ even}$

Step 4 of 5

(D) $f(t) = t^2 + t^4$

$f(-t) = t^2 + t^4$ $\boxed{f(t) is\ even}$

Step 5 of 5

(E) $f(t) = \sinh t$

$f(-t) = \sinh(-t) = -\sinh t.$

$\boxed{f(t) \textit{is even}}$